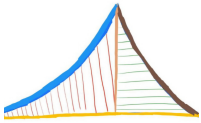
Matrices In Geometry



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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Contents

1 AE 2026 4
1.1 Formulae 4

2 BT 2025 16
2.1 Formulae 16

3 AG 2025 20
3.1 Formulae 20
3.2 Solved Questions 25

1. AE 2026

A. Formulae

1.1.1

$$\Pr(M) + \Pr(E) - \Pr(ME) + \Pr(M+E)' = 1 \tag{1.1.1.1}$$

1.1.2

$$A + A' = 1, \quad AA' = 0 \tag{1.1.2.1}$$

$$\implies \Pr(A + A') = \Pr(1) = 1 \quad \Pr(AA') = \Pr(0) = 0 \tag{1.1.2.2}$$

1.1.3

$$\Pr(A + A') = \Pr(A) + \Pr(A') \tag{1.1.3.1}$$

1.1.4 Difference rule of probability

$$\Pr(AB') = \Pr(A) - \Pr(AB) \tag{1.1.4.1}$$

Solution: From (1.1.2.1)

$$A = A \cdot 1 = A(B + B') = AB + AB' \tag{1.1.4.2}$$

Also,

$$(AB)(AB') = 0 \tag{1.1.4.3}$$

From (1.1.3.1),

$$\Pr(A) = \Pr(AB + AB') = \Pr(AB) + \Pr(AB') \tag{1.1.4.4}$$

Rearranging, we get:

$$P(AB') = P(A) - P(AB) \tag{1.1.4.5}$$

1.1.5 The Laplace transform is defined as

$$g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s) = \int_0^\infty e^{-st} g(t)\, dt \tag{1.1.5.1}$$

1.1.6

$$g'(t) \overset{\mathcal{L}}{\longleftrightarrow} sG(s) - g(0) \tag{1.1.6.1}$$

Solution: Applying integration by parts,

$$u = e^{-st}, dv = g'(t)\, dt \tag{1.1.6.2}$$

$$\implies \int_0^\infty e^{-st} g'(t)\, dt = \left[e^{-st} g(t)\right]_0^\infty - \int_0^\infty \left(-se^{-st}\right) g(t)\, dt \tag{1.1.6.3}$$

$$= -g(0) + s\int_0^\infty e^{-st} g(t)\, dt \tag{1.1.6.4}$$

yielding (1.1.6.1).

1.1.7

$$g''(t) \overset{\mathcal{L}}{\longleftrightarrow} s^2 G(s) - sg(0) - g'(0) \tag{1.1.7.1}$$

1.1.8

$$e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s-a) \tag{1.1.8.1}$$

Solution:

$$\implies e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty e^{-st} \left(e^{at} g(t)\right) dt \tag{1.1.8.2}$$

$$= \int_0^\infty e^{-(s-a)t} g(t)\, dt \tag{1.1.8.3}$$

yielding (1.1.8.1).

1.1.9 The unit step function is defined as

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} \tag{1.1.9.1}$$

1.1.10

$$u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s} \tag{1.1.10.1}$$

Solution:

$$\int_0^\infty e^{-st}\, dt = \left(-\frac{e^{-st}}{s}\right)_0^\infty = \frac{1}{s} \quad \Re(s) > 0 \tag{1.1.10.2}$$

1.1.11

$$tu(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s^2} \tag{1.1.11.1}$$

Solution: Differentiating (1.1.10.2) with respect to s results in (1.1.11.1).

$$\begin{aligned} \cos(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2+a^2} \\ \sin(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \end{aligned} \quad \Re(s) > a \tag{1.1.11.2}$$

Solution: From (1.1.8.1) and (1.1.10.1),

$$e^{\jmath at} u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s-\jmath a}, \quad \Re(s) > a \tag{1.1.11.3}$$

$$\implies \cos(at)u(t) + \jmath \sin(at)u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s+\jmath a}{(s-\jmath a)(s+\jmath a)} \tag{1.1.11.4}$$

$$= \frac{s}{s^2+a^2} + \jmath \frac{a}{s^2+a^2} \tag{1.1.11.5}$$

resulting in (1.1.11.2).

1.1.12 Solve

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \tag{1.1.12.1}$$

Solution: Applying the Laplace transform on both sides, and using (1.1.7.1),

$$s^2Y(s) - sy(0) - y'(0) = -\alpha^2 Y(s) \tag{1.1.12.2}$$

Substituting

$$y(0) = \beta, y'(0) = \gamma \tag{1.1.12.3}$$

in (1.1.12.2),

$$Y(s) = \frac{s\beta + \gamma}{s^2 + \alpha^2} \tag{1.1.12.4}$$

$$= \frac{s\beta}{s^2 + \alpha^2} + \frac{\gamma}{s^2 + \alpha^2} \tag{1.1.12.5}$$

yielding

$$y(t) = \beta \cos(\alpha t) + \frac{\gamma}{\alpha} \sin(\alpha t) \tag{1.1.12.6}$$

from (1.1.11.2).

1.1.13 (Euler's Method of Forward Differences.) Numerically solve

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0 \tag{1.1.13.1}$$

Solution: By definition,

$$\left.\frac{dy}{dx}\right|_{x=x_0} = \lim_{h \to 0} \frac{y(x_0 + h) - y(x_0)}{h} \tag{1.1.13.2}$$

which can be expressed using the notation

$$y_0' = \frac{y_1 - y_0}{h} \tag{1.1.13.3}$$

Similarly,

$$y_n'' = \frac{y_{n+1}' - y_n'}{h} \tag{1.1.13.4}$$

From (1.1.13.3),

$$y_{n+1}' = \frac{y_{n+2} - y_{n+1}}{h}, y_n' = \frac{y_{n+1} - y_n}{h} \tag{1.1.13.5}$$

yielding

$$y_n'' = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \tag{1.1.13.6}$$

upon simplification.

1.1.14 Kronecker delta function

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.14.1}$$

1.1.15

$$f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi n f_0 t}\, dt = \delta_n \tag{1.1.15.1}$$

1.1.16 The Fourier Series. Let

$$x(t) = \sum_{k=-\infty}^{\infty} x_0(t)\,(t - nT_0) \tag{1.1.16.1}$$

be a periodic signal with

$$x(t) = \begin{cases} x_0(t), & t \in \left(-\frac{T_0}{2}, \frac{T_0}{2}\right) \\ 0 & \text{otherwise} \end{cases}. \tag{1.1.16.2}$$

Then,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{\mathrm{j}2\pi n f_0 t}, \quad f_0 = \frac{1}{T_0} \tag{1.1.16.3}$$

Show that

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.16.4}$$

Solution: From (1.1.16.3),

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t} \right) e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.16.5}$$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi (k-n) f_0 t}\, dt \tag{1.1.16.6}$$

$$= \frac{1}{f_0} \sum_{k=-\infty}^{\infty} c_k \delta_{k-n} \tag{1.1.16.7}$$

upon substituting from (1.1.16.2) and (1.1.15.1), yielding (1.1.16.4).

1.1.17 The rectangular function. The unit rectangular function is defined as

$$\text{rect}\,(t) = \begin{cases} 1 & t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \tag{1.1.17.1}$$

1.1.18 Square Wave. Consider the Fourier series with

$$x_0(t) = \text{rect}\,(t), \quad T_0 = 2. \tag{1.1.18.1}$$

Show that

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} \tag{1.1.18.2}$$

Solution: Substituting $f_0 = \frac{1}{2}$ in (1.1.16.4),

$$c_n = \frac{1}{2}\int_{-1}^{1} e^{-j\pi kt}\,dt \tag{1.1.18.3}$$

$$= \left[\frac{\sin(\pi kt)}{2\pi k}\right]_{-\frac{1}{2}}^{\frac{1}{2}} \tag{1.1.18.4}$$

The following code verifies the result in Fig. 1.1.18.1. Note that Fig. 1.1.18.1 is continuous while (1.1.18.1) is discontinuous. So (1.1.18.2) provides only an approximation. This is known as the Gibbs' phenomenon.

codes/Fourier.py

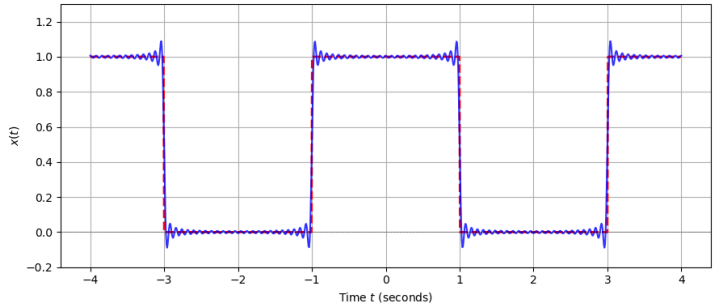


Fig. 1.1.18.1. Gibbs phenomenon

1.1.19 If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} c_n, \tag{1.1.19.1}$$

show that

$$x(t-t_0) \overset{\mathcal{F}}{\longleftrightarrow} c_n e^{-\mathrm{J}2\pi nt_0}, \tag{1.1.19.2}$$

Solution:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t-t_0)\,e^{-\mathrm{J}2\pi nf_0t}\,dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\,e^{-\mathrm{J}2\pi nf_0(t+t_0)}\,dt \tag{1.1.19.3}$$

$$= e^{\mathrm{J}2\pi nf_0t_0}\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\,e^{-\mathrm{J}2\pi nf_0t}\,dt \tag{1.1.19.4}$$

after a change of variables. Note that the limits of integration are unchanged due the periodicity of $x(t)$. Transferring the exponential term in (1.1.19.4) to the L.H.S yields (1.1.19.2).

1.1.20 The standard 2-D basis vectors are defined as

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.1.20.1}$$

Also,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.1.20.2}$$

1.1.21 Find the echelon form of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.1.21.1}$$

Solution: Following row reduction,

$$\mathbf{A} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{1.1.21.2}$$

1.1.22 The rank of a matrix is defined to be the number of nonzero rows in echelon form. $\therefore \text{rank}\,(\mathbf{A}) = 1$.

1.1.23 A matrix is said to be singular if its rank is nonzero.

1.1.24 For

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top, \text{rank}\,(\mathbf{A}) = 1. \tag{1.1.24.1}$$

where both $\mathbf{u}$ and $\mathbf{v}$ are n-dimensional real-valued non-zero column vectors.

Solution: Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}. \tag{1.1.24.2}$$

Then

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \tag{1.1.24.3}$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.1.24.4}$$

From (1.1.21.2) and Definition 1.1.22, we obtain (1.1.24.1).

1.1.25

$$\left|\mathbf{A}^\top\right| = \left|\mathbf{A}\right| \tag{1.1.25.1}$$

1.1.26

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right| \tag{1.1.26.1}$$

Solution: If $\mathbf{A}$ is invertible

$$\mathbf{A} = \mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_k \tag{1.1.26.2}$$

where $\mathbf{E}_i$ are invertible matrices. For any single elementary matrix $\mathbf{E}$, it is easy to verify that

$$\left|\mathbf{EB}\right| = \left|\mathbf{E}\right|\left|\mathbf{B}\right|. \tag{1.1.26.3}$$

$$\therefore \left|\mathbf{AB}\right| = \left|\mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_k\mathbf{B}\right| \tag{1.1.26.4}$$

$$= \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\cdots\mathbf{E}_k\mathbf{B}\right| \tag{1.1.26.5}$$

$$= \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\right|\cdots\left|\mathbf{E}_k\right|\left|\mathbf{B}\right| \tag{1.1.26.6}$$

For $\mathbf{B} = \mathbf{I}$,

$$\left|\mathbf{A}\right| = \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\right|\cdots\left|\mathbf{E}_k\right| \tag{1.1.26.7}$$

which upon substituting in (1.1.26.6) yields

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right| \tag{1.1.26.8}$$

1.1.27

$$\left|\mathbf{A^{-1}}\right| = \frac{1}{\left|\mathbf{A}\right|} \tag{1.1.27.1}$$

1.1.28 If

$$\mathbf{Ax} = \lambda\mathbf{x}, \tag{1.1.28.1}$$

then λ is an eigenvalue $\mathbf{x}$ of $\mathbf{A}$.

1.1.29 Let $\lambda_1, \lambda_2, ..., \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then,

$$\sum_{i=1}^{n}\lambda_i = \mathrm{Tr}\left(\mathbf{A}\right) \tag{1.1.29.1}$$

$$\prod_{i=1}^{n}\lambda_i = \left|\mathbf{A}\right| \tag{1.1.29.2}$$

1.1.30 $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.

Solution: From (1.1.27.1),

$$\left|\mathbf{A} - \lambda\mathbf{I}\right| = \left|\left(\mathbf{A} - \lambda\mathbf{I}\right)^\top\right| \tag{1.1.30.1}$$

$$= \left|\mathbf{A}^\top - \lambda\mathbf{I}\right|, \tag{1.1.30.2}$$

completing the proof.

1.1.31 The equation of a conic with directrix $\mathbf{n}^\top\mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given

by

$$\mathrm{g}\left(\mathbf{x}\right)=\mathbf{x}^{\top}\mathbf{V}\mathbf{x}+2\mathbf{u}^{\top}\mathbf{x}+f=0 \tag{1.1.31.1}$$

where

$$\mathbf{V}=\left\|\mathbf{n}\right\|^{2}\mathbf{I}-e^{2}\mathbf{n}\mathbf{n}^{\top}, \tag{1.1.31.2}$$

$$\mathbf{u}=ce^{2}\mathbf{n}-\left\|\mathbf{n}\right\|^{2}\mathbf{F}, \tag{1.1.31.3}$$

$$f=\left\|\mathbf{n}\right\|^{2}\left\|\mathbf{F}\right\|^{2}-c^{2}e^{2} \tag{1.1.31.4}$$

Solution:

$$\left\|\mathbf{x}-\mathbf{F}\right\|^{2}=e^{2}\frac{(\mathbf{n}^{\top}\mathbf{x}-c)^{2}}{\left\|\mathbf{n}\right\|^{2}} \tag{1.1.31.5}$$

$$\implies \left\|\mathbf{n}\right\|^{2}(\mathbf{x}-\mathbf{F})^{\top}(\mathbf{x}-\mathbf{F})=e^{2}\left(\mathbf{n}^{\top}\mathbf{x}-c\right)^{2} \tag{1.1.31.6}$$

$$\implies \left\|\mathbf{n}\right\|^{2}\left(\mathbf{x}^{\top}\mathbf{x}-2\mathbf{F}^{\top}\mathbf{x}+\left\|\mathbf{F}\right\|^{2}\right)=e^{2}\left(c^{2}+\left(\mathbf{n}^{\top}\mathbf{x}\right)^{2}-2c\mathbf{n}^{\top}\mathbf{x}\right) \tag{1.1.31.7}$$

$$=e^{2}\left(c^{2}+\left(\mathbf{x}^{\top}\mathbf{n}\mathbf{n}^{\top}\mathbf{x}\right)-2c\mathbf{n}^{\top}\mathbf{x}\right) \tag{1.1.31.8}$$

which can be expressed as (1.1.31.1) after simplification.

1.1.32 The polar form of a conic section is given as:

$$r=\frac{ep}{1+e\cos\left(\theta-\theta_{0}\right)} \tag{1.1.32.1}$$

where

- r is the distance from the focus(origin) to a point on the curve.
- e is the eccentricity of the conic
- p is the distance from the focus to the directrix
- θ_0 is the angle that defines the orientation of the major axis of the conic section relative to the polar reference axis ($\theta = 0$).

Solution: From (1.1.31.1), the general equation of a conic in matrix form is given by

$$\mathbf{x}^{\top}\mathbf{V}\mathbf{x}+2\mathbf{u}^{\top}\mathbf{x}+f=0 \tag{1.1.32.2}$$

For a conic with a focus at the origin ($\mathbf{F}=\mathbf{0}$), unit normal vector $\mathbf{n}=\begin{pmatrix}\cos\theta_{0}\\ \sin\theta_{0}\end{pmatrix}$, eccentricity e, and directrix distance parameter $c=-p$, substituting into (1.1.31.2)–(1.1.31.4) yields:

$$\mathbf{V}=\mathbf{I}-e^{2}\mathbf{n}\mathbf{n}^{\top}, \tag{1.1.32.3}$$

$$\mathbf{u}=-pe^{2}\mathbf{n}, \tag{1.1.32.4}$$

$$f=-p^{2}e^{2} \tag{1.1.32.5}$$

Expressing $\mathbf{x}$ directly in polar coordinates yields

$$\mathbf{x}=r\mathbf{r}\left(\theta\right),\quad \text{where } \mathbf{r}\left(\theta\right)=\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix} \tag{1.1.32.6}$$

Substituting into (1.1.32.2) yields

$$r^2 \left(\mathbf{r}(\theta)^\top \mathbf{V}\mathbf{r}(\theta) \right) + 2r \left(\mathbf{u}^\top \mathbf{r}(\theta) \right) + f = 0 \tag{1.1.32.7}$$

Evaluating the inner products:

$$\mathbf{n}^\top \mathbf{r}(\theta) = \begin{pmatrix} \cos\theta_0 & \sin\theta_0 \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} = \cos(\theta - \theta_0) \tag{1.1.32.8}$$

$$\mathbf{r}(\theta)^\top \mathbf{V}\mathbf{r}(\theta) = \mathbf{r}(\theta)^\top \left(\mathbf{I} - e^2 \mathbf{n}\mathbf{n}^\top \right) \mathbf{r}(\theta) = 1 - e^2 \cos^2(\theta - \theta_0) \tag{1.1.32.9}$$

$$\mathbf{u}^\top \mathbf{r}(\theta) = -pe^2 \cos(\theta - \theta_0) \tag{1.1.32.10}$$

Substituting these into (1.1.32.7):

$$r^2 \left[1 - e^2 \cos^2(\theta - \theta_0) \right] - 2rpe^2 \cos(\theta - \theta_0) - p^2 e^2 = 0 \tag{1.1.32.11}$$

Factoring the quadratic expression:

$$\left[r(1 + e\cos(\theta - \theta_0)) - ep \right] \left[r(1 - e\cos(\theta - \theta_0)) + ep \right] = 0 \tag{1.1.32.12}$$

Taking the positive real root for radial distance $r > 0$:

$$r(1 + e\cos(\theta - \theta_0)) = ep \tag{1.1.32.13}$$

yielding (1.1.32.1)

1.1.33 Gravitational parameter of a body μ is defined as

$$\mu = GM \tag{1.1.33.1}$$

where M is the mass of the body

1.1.34 Specific angular momentum for a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$ is

$$h = \|\mathbf{r} \times \mathbf{v}\| \tag{1.1.34.1}$$

1.1.35 Specific Energy for a satellite with position vector $\mathbf{r}$, velocity vector $\mathbf{v}$ and gravitational parameter μ is

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} \tag{1.1.35.1}$$

where

$$r = \|\mathbf{r}\|, v = \|\mathbf{v}\| \tag{1.1.35.2}$$

1.1.36 The gravitational acceleration $\mathbf{a}$ acting on a satellite is given by Newton's law of gravitation:

$$\mathbf{a} = -\frac{\mu}{r^3}\mathbf{r} \tag{1.1.36.1}$$

1.1.37 The trajectory of a satellite under a central gravitational force is always a conic section.

Solution: Taking the cross product of $\mathbf{r}$ with $\mathbf{a}$:

$$\mathbf{r} \times \mathbf{a} = \mathbf{r} \times \left(-\frac{\mu}{r^3}\mathbf{r}\right) = \mathbf{0} \tag{1.1.37.1}$$

From (1.1.37.1)

$$\frac{d}{dt}\left(\mathbf{r} \times \mathbf{v}\right) = \mathbf{v} \times \mathbf{v} + \mathbf{r} \times \mathbf{a} = \mathbf{0} \tag{1.1.37.2}$$

Hence, the specific angular momentum vector $\mathbf{h}$ is conserved:

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = \text{constant} \tag{1.1.37.3}$$

Crossing the acceleration $\mathbf{a}$ with $\mathbf{h}$:

$$\mathbf{a} \times \mathbf{h} = \left(-\frac{\mu}{r^3}\mathbf{r}\right) \times \left(\mathbf{r} \times \mathbf{v}\right) \tag{1.1.37.4}$$

$$= -\frac{\mu}{r^3}\left[\left(\mathbf{r} \cdot \mathbf{v}\right)\mathbf{r} - \left(\mathbf{r} \cdot \mathbf{r}\right)\mathbf{v}\right] \tag{1.1.37.5}$$

$$= \frac{\mu}{r^3}\left[r^2\mathbf{v} - \left(\mathbf{r} \cdot \mathbf{v}\right)\mathbf{r}\right] \tag{1.1.37.6}$$

Using $\mathbf{r} \cdot \mathbf{v} = r\frac{dr}{dt}$, this simplifies to:

$$\mathbf{a} \times \mathbf{h} = \mu\left[\frac{\mathbf{v}}{r} - \frac{1}{r^2}\frac{dr}{dt}\mathbf{r}\right] = \mu\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right) \tag{1.1.37.7}$$

Integrating with respect to time t:

$$\mathbf{v} \times \mathbf{h} = \mu\left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.37.8}$$

where $\mathbf{C}$ is the integration constant. Taking the scalar product of (1.1.37.8) with $\mathbf{r}$:

$$\mathbf{r} \cdot \left(\mathbf{v} \times \mathbf{h}\right) = \mu\mathbf{r} \cdot \left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.37.9}$$

$$\implies \left(\mathbf{r} \times \mathbf{v}\right) \cdot \mathbf{h} = \mu\left(r + rC\cos\theta\right) \tag{1.1.37.10}$$

$$\implies \left\|\mathbf{h}\right\|^2 = \mu r\left(1 + C\cos\theta\right) \tag{1.1.37.11}$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{C}$. Rearranging for r yields

$$r = \frac{\frac{\left\|\mathbf{h}\right\|^2}{\mu}}{1 + C\cos\theta} = \frac{ep}{1 + e\cos\theta} \tag{1.1.37.12}$$

This is the same equation as (1.1.32.1) where

$$ep = \frac{\left\|\mathbf{h}\right\|^2}{\mu} \tag{1.1.37.13}$$

$$\mathbf{e} = \mathbf{C} \tag{1.1.37.14}$$

1.1.38 The squared eccentricity e^2 of a satellite's trajectory is related to its specific orbital

energy ε and specific angular momentum magnitude h by

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.38.1}$$

Solution: From (1.1.37.8) and (1.1.37.14),

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \tag{1.1.38.2}$$

Taking the magnitude squared of both sides using $\|\mathbf{e}\|^2 = \mathbf{e} \cdot \mathbf{e}$:

$$e^2 = \|\mathbf{e}\|^2 = \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \cdot \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \tag{1.1.38.3}$$

Expanding the inner product:

$$e^2 = \frac{\|\mathbf{v} \times \mathbf{h}\|^2}{\mu^2} - \frac{2}{\mu r}\left[\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h})\right] + \frac{\mathbf{r} \cdot \mathbf{r}}{r^2} \tag{1.1.38.4}$$

We know that

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mathbf{h} \cdot \mathbf{h} = h^2 \tag{1.1.38.5}$$

And,

$$\mathbf{r} \cdot \mathbf{r} = r^2 \tag{1.1.38.6}$$

Substituting these back into (1.1.38.4) yields

$$e^2 = \frac{v^2 h^2}{\mu^2} - \frac{2h^2}{\mu r} + 1 \tag{1.1.38.7}$$

$$= 1 + \frac{2h^2}{\mu^2}\left(\frac{v^2}{2} - \frac{\mu}{r}\right) \tag{1.1.38.8}$$

Substituting (1.1.35.1) into (1.1.38.8) yields:

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.38.9}$$

1.1.39 Consider the system A system comprising a bar, spring and mass is shown in the figure Fig. 1.1.39.1 The bar, having negligible mass, is made of a material having Young's modulus E, cross-sectional area A and length L. The spring stiffness is k and the mass is M kg. Find the axial stiffness of the bar.

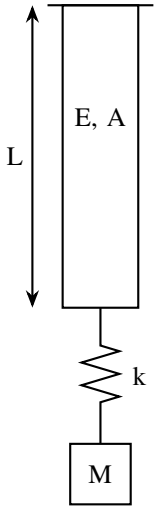


Fig. 1.1.39.1. Spring mass system

Solution: The axial stiffness of the bar is

$$k_b = \frac{AE}{L} \tag{1.1.39.1}$$

1.1.40 Find the overall spring constant in Fig. 1.1.39.1.

Solution: Using Hooke's law, by balancing the displacements,

$$\frac{Mg}{k_t} = \frac{Mg}{k_b} + \frac{Mg}{k} \tag{1.1.40.1}$$

$$\implies k_{eq} = \frac{k_b k}{k_b + k} \tag{1.1.40.2}$$

where k_t is the overall spring constant.

1.1.41 Show that the equation for motion for the mass in Fig. 1.1.41.1 is

$$\frac{d^2x}{dt^2} = -\frac{k}{M}x \tag{1.1.41.1}$$

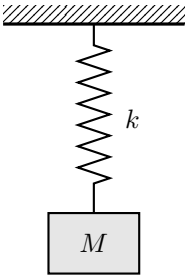


Fig. 1.1.41.1. Spring Mass system in vertical plane

Solution: When the mass M is at rest, let the system stretch by δ_{st}.Then,

$$Mg - k\delta_{st} = 0 \implies Mg = k\delta_{st} \tag{1.1.41.2}$$

When the mass is displaced downward by $x(t)$, using Hooke's law,

$$M\frac{d^2x}{dt^2} = Mg - k(\delta_{\text{st}} + x) \tag{1.1.41.3}$$

$$\tag{1.1.41.4}$$

yielding (1.1.41.1), upon substituting from (1.1.41.2).

2. BT 2025

A. Formulae

2.1.1 Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \tag{2.1.1.1}$$

2.1.2 The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{2.1.2.1}$$

2.1.3 The parametric form of the equation of a line is

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m} \tag{2.1.3.1}$$

where

$$\mathbf{m}^\top \mathbf{n} = 0 \tag{2.1.3.2}$$

2.1.4 A circle with center $\mathbf{c}$ and radius r, can be expressed as (1.1.31.1) with

$$\mathbf{V} = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = -\mathbf{c}, \quad f = \|\mathbf{c}\|^2 - r^2 \tag{2.1.4.1}$$

Solution: The equation of a circle is

$$\|\mathbf{x} - \mathbf{c}\|^2 = r^2 \tag{2.1.4.2}$$

$$(\mathbf{x} - \mathbf{c})^\top (\mathbf{x} - \mathbf{c}) = r^2 \tag{2.1.4.3}$$

$$\mathbf{x}^\top \mathbf{x} - 2\mathbf{c}^\top \mathbf{x} + \mathbf{c}^\top \mathbf{c} - r^2 = 0 \tag{2.1.4.4}$$

yielding (2.1.4.1) upon comparing with (1.1.31.1)

2.1.5 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by

$$\mathbf{X} = \begin{pmatrix} \mathbf{h} & \mathbf{m} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ \kappa_1 & \kappa_2 \end{pmatrix} \tag{2.1.5.1}$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V}\mathbf{h}+\mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top (\mathbf{V}\mathbf{h}+\mathbf{u})\right]^2 - g(\mathbf{h})\left(\mathbf{m}^\top \mathbf{V}\mathbf{m}\right)}}{\mathbf{m}^\top \mathbf{V}\mathbf{m}}, i = 1, 2. \tag{2.1.5.2}$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$.

2.1.6 (Intersection of Conics and Common Chord) The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ is given by the pencil

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu\mathbf{V}_2)\,\mathbf{x} + 2\left(\mathbf{u}_1 + \mu\mathbf{u}_2\right)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \tag{2.1.6.1}$$

The value of μ corresponding to the degenerate conic (the radical axis / common chord line) is found from

$$\begin{vmatrix} \mathbf{V}_1 + \mu\mathbf{V}_2 & \mathbf{u}_1 + \mu\mathbf{u}_2 \\ (\mathbf{u}_1 + \mu\mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \tag{2.1.6.2}$$

2.1.7 For a differentiable function $f(x)$,

$$f''(x) > 0 \quad \forall x \in (0, \infty) \implies f(x) \text{ is strictly convex} \tag{2.1.7.1}$$

2.1.8 (Gradient Descent Optimization) For the convex optimization problem

$$\min_x f(x), \tag{2.1.8.1}$$

$$x_{k+1} = x_k - \eta f'(x_k) \tag{2.1.8.2}$$

is the iterative parameter update with learning rate $\eta > 0$.

2.1.9 (Newton-Raphson Method) For a real-valued differentiable function $f(x)$, the root is iteratively approximated using its first-order Taylor series expansion about the current estimate x_n

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{2.1.9.1}$$

2.1.10 (System of Linear Equations) The matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{2.1.10.1}$$

is consistent iff

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \tag{2.1.10.2}$$

2.1.11 Sum of Independent Discrete Random Variables (Two Dice Roll)

- Let $X_1, X_2 \in \{1, 2, 3, 4, 5, 6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF)

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{6}, & 1 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.11.1}$$

- (Convolution) Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by

the discrete convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$:

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k \Pr(X_1 = n-k \mid X_2 = k)\, p_{X_2}(k) \tag{2.1.11.2}$$

$$= \sum_k p_{X_1}(n-k)\, p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \tag{2.1.11.3}$$

- Substituting (2.1.11.1) into (2.1.11.3) yields the triangular PMF

$$p_X(n) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \le n \le 7 \\ \frac{13-n}{36}, & 7 < n \le 12 \\ 0, & n > 12 \end{cases} \tag{2.1.11.4}$$

The following code verifies (2.1.11.4) in Fig. 2.1.11.1.

codes/btq35/dice.py

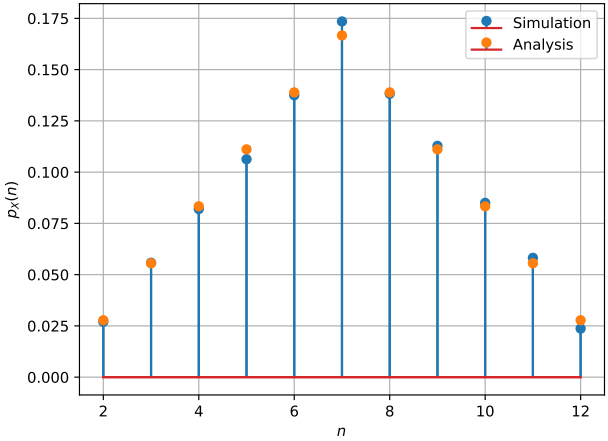


Fig. 2.1.11.1. PMF of the sum of two fair dice: Simulation vs Analysis

2.1.12 (Unit Step Response of an RC Circuit) Consider a series RC circuit excited by a step input $v_i(t) = V_0 u(t)$ where V_0 is a constant. Applying KVL around the loop,

$$RC\frac{dv_o}{dt} + v_o(t) = V_0 u(t) \tag{2.1.12.1}$$

yielding

$$\frac{V_0}{s} = sCRV_o(s) + V_o(s) \tag{2.1.12.2}$$

$$\implies V_o(s) = \frac{V_0}{s(RCs+1)} \tag{2.1.12.3}$$

$$= V_0\left[\frac{1}{s} - \frac{1}{s+\frac{1}{RC}}\right] \tag{2.1.12.4}$$

$$\implies v_o(t) = V_0\left(1 - e^{-\frac{t}{RC}}\right)u(t) \tag{2.1.12.5}$$

2.1.13 (Continuity) A real-valued function $f(x)$ is continuous at $x = x_0$ iff

$$\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x) = f(x_0) \tag{2.1.13.1}$$

2.1.14 (Differentiability) A function $f(x)$ is differentiable at $x = x_0$ if

$$f'\left(x_0^-\right) = \lim_{h\to 0^-} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{h\to 0^+} \frac{f(x_0+h) - f(x_0)}{h} = f'\left(x_0^+\right) \tag{2.1.14.1}$$

2.1.15 If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$

$$\lim_{x\to x_0}\left[f(x) - f(x_0)\right] = \lim_{x\to x_0}\left[\frac{f(x) - f(x_0)}{x - x_0}(x - x_0)\right] = f'(x_0)\cdot 0 = 0 \tag{2.1.15.1}$$

2.1.16 Piecewise Boundary Conditions at $x = a$

- Consider a function $f(x)$ defined by differentiable branches $f_1(x)$ and $f_2(x)$ across a boundary point $x = a$:

$$f(x) = \begin{cases} f_1(x), & x \le a \\ f_2(x), & x > a \end{cases} \tag{2.1.16.1}$$

For $f(x)$ to be continuous at $x = a$, the one-sided limits must equal the boundary value:

$$\lim_{x\to a^-} f_1(x) = \lim_{x\to a^+} f_2(x) \implies f_1(a) = f_2(a) \tag{2.1.16.2}$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives at the boundary must coincide:

$$f_1'(a) = f_2'(a) \tag{2.1.16.3}$$

2.1.17 The Z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n} \tag{2.1.17.1}$$

over the Region of Convergence (ROC) where the series converges absolutely.

2.1.18 The discrete unit step function is defined as

$$u\left(n\right)=\begin{cases}1, & n\geq 0\\ 0, & n<0\end{cases} \tag{2.1.18.1}$$

and

$$u\left[n\right]\overset{\mathcal{Z}}{\longleftrightarrow}\sum_{n=0}^{\infty}z^{-n}=\frac{1}{1-z^{-1}},\quad |z|>1 \tag{2.1.18.2}$$

2.1.19 Time-Shift Property

$$x\left(n-k\right)\overset{\mathcal{Z}}{\longleftrightarrow}z^{-k}X\left(z\right) \tag{2.1.19.1}$$

2.1.20 Standard Inverse Transform Pair: For any scalar constant α,

$$\alpha^{n}u\left(n\right)\overset{\mathcal{Z}}{\longleftrightarrow}\frac{1}{1-\alpha z^{-1}},\quad |z|>|\alpha| \tag{2.1.20.1}$$

2.1.21 (Trace of a Matrix) Let $\mathbf{A}\in\mathbb{R}^{n\times n}$ be a square matrix with eigenvalues $\lambda_1,\lambda_2,\ldots,\lambda_n$. Then,

$$\sum_{i=1}^{n}\lambda_i=\sum_{i=1}^{n}a_{ii}=\mathrm{tr}\left(\mathbf{A}\right) \tag{2.1.21.1}$$

Solution: The characteristic polynomial of $\mathbf{A}$ is defined by

$$p_{\mathbf{A}}\left(\lambda\right)=\left|\lambda\mathbf{I}-\mathbf{A}\right| \tag{2.1.21.2}$$

$$=\prod_{i=1}^{n}\left(\lambda-\lambda_i\right)=\lambda^{n}-\left(\sum_{i=1}^{n}\lambda_i\right)\lambda^{n-1}+\cdots+\left(-1\right)^{n}\prod_{i=1}^{n}\lambda_i \tag{2.1.21.3}$$

Evaluating the determinant $\left|\lambda\mathbf{I}-\mathbf{A}\right|$ using cofactor expansion, the only term that contributes to λ^{n} and λ^{n-1} is the product of the principal diagonal entries

$$\prod_{i=1}^{n}\left(\lambda-a_{ii}\right)=\lambda^{n}-\left(\sum_{i=1}^{n}a_{ii}\right)\lambda^{n-1}+\ldots \tag{2.1.21.4}$$

where a_{ii} denotes the diagonal elements of $\mathbf{A}$. Equating the coefficient of λ^{n-1} from (2.1.21.3) and (2.1.21.4)

3. AG 2025

A. Formulae

3.1.1 (Bernoulli Distribution) For $X\in\{0,1\}$, the pmf is defined as

$$P_X(k)=\begin{cases}p & k=1\\ q & k=0\\ 0 & \text{otherwise}\end{cases} \tag{3.1.1.1}$$

3.1.2 The moment generating function

$$\begin{aligned} M_X(z) \triangleq E\left[z^{-X}\right] &= p_X(1)z^{-1} + p_X(0)z^0 \\ &= pz^{-1} + q \end{aligned} \tag{3.1.2.1}$$

upon substituting from (2.1.17.1).

3.1.3 (Binomial Distribution) If X_i are independent, for

$$\begin{aligned} Y \triangleq \sum_{i=1}^{n} X_i, \quad M_Y(z) &= \prod_{i=1}^{n} M_{X_i}(z) \\ &= \left(pZ^{-1} + q\right)^n \end{aligned} \tag{3.1.3.1}$$

Solution: From (3.1.2.1),

$$M_Y(z) = E\left[z^{-\sum_{i=1}^{n} X_i}\right] = E\left[\prod_{i=1}^{n} z^{-X_i}\right] \tag{3.1.3.2}$$

$$= \prod_{i=1}^{n} E\left[z^{-X_i}\right] \tag{3.1.3.3}$$

yielding (3.1.3.1).

3.1.4 From (3.1.3.1),

$$\left(pz^{-1} + q\right)^n = {}^nC_k p^k q^{n-k} z^{-k} \tag{3.1.4.1}$$

$$\implies p_Y(k) = \begin{cases} {}^nC_k p^k q^{n-k} & 0 \le k \le n \\ 0 & \text{otherwise} \end{cases} \tag{3.1.4.2}$$

using the definition in (3.1.2.1) along with (2.1.17.1).

3.1.5 Laplace transforms of some common functions:

3.1.6

$$\cos(at) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2 + a^2} \tag{3.1.6.1}$$

Proof:

$$\cos(at) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty \cos(at)\ e^{-st}\, dt = I \tag{3.1.6.2}$$

$$I = \int_0^\infty \frac{d(\sin(at))}{a}\, e^{-st} \tag{3.1.6.3}$$

$$= \frac{\sin(at)}{a}\, e^{-st}\Big|_0^\infty - \frac{1}{a}\int_0^\infty \sin(at) e^{-st}\,(-s)\, dt \tag{3.1.6.4}$$

$$= 0 + \frac{s}{a}\int_0^\infty \frac{d(-\cos(at))}{a}\, e^{-st} \tag{3.1.6.5}$$

$$= -\left(\frac{s}{a^2}\right)\left\{\cos(at)\ e^{-st}\Big|_0^\infty - \int_0^\infty \cos(at)\ e^{-st}\ (-s)\right\} \tag{3.1.6.6}$$

$$= -\left(\frac{s}{a^2}\right)\{(0-1) - (-s)\, I\} \tag{3.1.6.7}$$

$$= -\left(\frac{s}{a^2}\right)\{-1 + sI\} \tag{3.1.6.8}$$

$$I = \frac{s}{a^2} - \frac{s^2}{a^2}(I) \tag{3.1.6.9}$$

$$\left(\frac{a^2+s^2}{a^2}\right) I = \frac{s}{a^2} \tag{3.1.6.10}$$

$$I = \frac{s}{s^2+a^2} \tag{3.1.6.11}$$

3.1.7

$$\sin(at) \overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \tag{3.1.7.1}$$

Proof:

$$\sin(at) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty \sin(at)\, e^{-st}\, dt = I \tag{3.1.7.2}$$

$$I = \int_0^\infty \frac{d(-\cos(at))}{a}\, e^{-st} \tag{3.1.7.3}$$

$$= \frac{-\cos(at)}{a}\, e^{-st}\Big|_0^\infty - \frac{1}{a}\int_0^\infty -\cos(at)e^{-st}(-s)dt \tag{3.1.7.4}$$

$$= \frac{1}{a} - \frac{s}{a}\int_0^\infty \frac{d(\sin(at))}{a}\, e^{-st} \tag{3.1.7.5}$$

$$= \frac{1}{a} - (\frac{s}{a^2})\left\{\sin(at)\, e^{-st}\,\Big|_0^\infty - \int_0^\infty \sin(at)\, e^{-st}\, (-s)\right\} \tag{3.1.7.6}$$

$$= -\left(\frac{s}{a^2}\right)\{(0-0) - (-s)\, I\} \tag{3.1.7.7}$$

$$= -\left(\frac{s}{a^2}\right)\{0 + sI\} \tag{3.1.7.8}$$

$$I = \frac{a}{a^2} - \frac{s^2}{a^2}\,(I) \tag{3.1.7.9}$$

$$\left(\frac{a^2+s^2}{a^2}\right) I = \frac{a}{a^2} \tag{3.1.7.10}$$

$$I = \frac{a}{s^2+a^2} \tag{3.1.7.11}$$

3.1.8

$$\sinh(at) \overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2-a^2} \tag{3.1.8.1}$$

Proof:

$$\sinh(at) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty \frac{e^{at}-e^{-at}}{2}\, e^{-st} dt \tag{3.1.8.2}$$

$$= \frac{1}{2}\int_0^\infty e^{(a-s)(t)} dt - \frac{1}{2}\int_0^\infty e^{(-a-s)(t)} dt \tag{3.1.8.3}$$

$$= \frac{1}{2}\left\{\frac{e^{(a-s)(t)}}{a-s}\right\}\Big|_0^\infty - \frac{1}{2}\left\{\frac{e^{(-a-s)(t)}}{-a-s}\right\}\Big|_0^\infty$$

(assume s>a for integral to be finite)

$$= \frac{1}{2}\left(\frac{1}{s-a} - \frac{1}{s+a}\right) \tag{3.1.8.4}$$

$$= \frac{1}{2}\left(\frac{2a}{s^2-a^2}\right) \tag{3.1.8.5}$$

$$I = \frac{a}{s^2-a^2} \tag{3.1.8.6}$$

3.1.9

$$\cosh(at) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2 - a^2} \tag{3.1.9.1}$$

Proof:

$$\cosh(at) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty \frac{e^{at} + e^{-at}}{2} e^{-st} dt \tag{3.1.9.2}$$

$$= \frac{1}{2}\int_0^\infty e^{(a-s)(t)} dt + \frac{1}{2}\int_0^\infty e^{(-a-s)(t)} dt \tag{3.1.9.3}$$

$$= \frac{1}{2}\left\{\frac{e^{(a-s)(t)}}{a-s}\right\}\Bigg|_0^\infty + \frac{1}{2}\left\{\frac{e^{(-a-s)(t)}}{-a-s}\right\}\Bigg|_0^\infty$$

(assume s>a for integral to be finite)

$$= \frac{1}{2}\left(\frac{1}{s-a} + \frac{1}{s+a}\right) \tag{3.1.9.4}$$

$$= \frac{1}{2}\left(\frac{2s}{s^2 - a^2}\right) \tag{3.1.9.5}$$

$$I = \frac{s}{s^2 - a^2} \tag{3.1.9.6}$$

3.1.10 Sum of n terms of a gp

$$\sum_{i=1}^{n} q^i = q\frac{q^n - 1}{q - 1} \tag{3.1.10.1}$$

3.1.11 Expansions:

$$\sin(x) = \sum_{i=1}^{n} -1^{i+1}\frac{x^{2i-1}}{(2i-1)!} \tag{3.1.11.1}$$

$$\cos(x) = \sum_{i=1}^{n} -1^{i+1}\frac{x^{i-1}}{i-1!} \tag{3.1.11.2}$$

3.1.12 Euler's Theorem for homogeneous functions:
It states that if a function $f(x, y)$is homogeneous of degree n,then the following condition holds:

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nf \tag{3.1.12.1}$$

3.1.13

$$f'(x) \overset{\mathcal{L}}{\longleftrightarrow} sG(s) - f(0) \tag{3.1.13.1}$$

Proof
Let G(s) be the laplace transform of y=f(x):

$$y \overset{\mathcal{L}}{\longleftrightarrow} G(s) \tag{3.1.13.2}$$

By definition of laplace transform,

$$G(s) = \int_0^\infty f(x) e^{-sx} dx \tag{3.1.13.3}$$

integrating by parts,

$$G(s) = \int_0^\infty f(x)\, d\left(\frac{e^{-sx}}{-s}\right) \tag{3.1.13.4}$$

$$= f(x)\, \frac{e^{-sx}}{-s}\bigg|_0^\infty + \int_0^\infty f'(x)\, \frac{e^{-sx}}{s} \tag{3.1.13.5}$$

$$= \left(\frac{f(\infty)\, e^{-\infty} - f(0)}{-s}\right) + \frac{1}{s}\, \mathcal{L}\{f'(x)\} \tag{3.1.13.6}$$

$$sG(s) = f(0) + \mathcal{L}\{f'(x)\} \tag{3.1.13.7}$$

$$f'(x) \overset{\mathcal{L}}{\longleftrightarrow} sG(s) - f(0) \tag{3.1.13.8}$$

3.1.14 The laplace transform is a linear function

3.1.15 The inverse laplace transform is a linear function

3.1.16 laplace transform of e^{ax}

$$e^{ax} \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s-a} \tag{3.1.16.1}$$

Proof:

$$e^{ax} \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty e^{at}\, e^{-st} dt \tag{3.1.16.2}$$

$$= \int_0^\infty e^{(a-s)(t)} dt \tag{3.1.16.3}$$

$$= \frac{1}{a-s}\, (e^{(a-s)(t)}\big|_0^\infty) \tag{3.1.16.4}$$

$$= \frac{1}{a-s}(0-1) \qquad \text{(Assuming that s>1)}$$

$$= \frac{1}{s-a} \tag{3.1.16.5}$$

3.1.17 Euler's forward difference method: The output of f(x) at $x = x_0$ is y_0, then,
$y_1 = y_0 + hf'(x_0, y_0)$
where y_1 is the value of the function at $x = x_0 + h$

B. Solved Questions

3.2.1 A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once? (AG 2025)

a) $\frac{75}{216}$ b) $\frac{1}{6}$ c) $\frac{1}{18}$ d) $\frac{25}{216}$

Solution: From (2.1.11.1),

$$P_{X_1}(k) = \begin{cases} \frac{1}{6} & k = 1 \\ \frac{5}{6} & k = 0 \end{cases} \tag{3.2.1.1}$$

where X_i represents the ith roll. Substituting $n = 3, p = \frac{1}{6}, q = \frac{5}{6}$, in (3.1.4.2), the desired probability is obtained as

$${}^3C_1 \frac{1}{6} \left(\frac{5}{6}\right)^2 = \frac{75}{216}. \tag{3.2.1.2}$$

The following code simulates the result

codes/Probability.c

3.2.2 If a, b, and c are the roots of

$$2x^3 - 3x^2 + px - 1 = 0 \tag{3.2.2.1}$$

and sum of the two roots is 1, the value of p is (AG 2025)

a) 2 b) 1 c) 3 d) $\frac{1}{3}$

Solution: Using the sum and product of roots from high school algebra and the given information,

$$a + b = 1 \tag{3.2.2.2}$$

$$a + b + c = \frac{3}{2} \tag{3.2.2.3}$$

$$abc = \frac{1}{2} \tag{3.2.2.4}$$

$$ab + bc + ac = \frac{p}{2} \tag{3.2.2.5}$$

From (3.2.2.2) and (3.2.2.3),

$$c = \frac{1}{2}, \tag{3.2.2.6}$$

(3.2.2.5) can be expressed as

$$\frac{p}{2} = \frac{abc}{c} + c\,(a + b) = \frac{1}{2c} + c \tag{3.2.2.7}$$

$$\implies p = 3 \tag{3.2.2.8}$$

upon substituting from (3.2.2.2), (3.2.2.4) and (3.2.2.6). Using the Newton-Raphson method the difference equation for computing the root of (3.2.2.1) is

$$x_{n+1} = x_n + \frac{2x^3 - 3x^2 + 3x - 1}{6x^2 - 6x + 3} \tag{3.2.2.9}$$

from (2.1.9.1). With an initial guess of 4, running this difference equation 6 times gives us the real root, which is $\frac{1}{2}$. The following code verifies the result in Fig. 3.2.2.1.

codes/raph.py

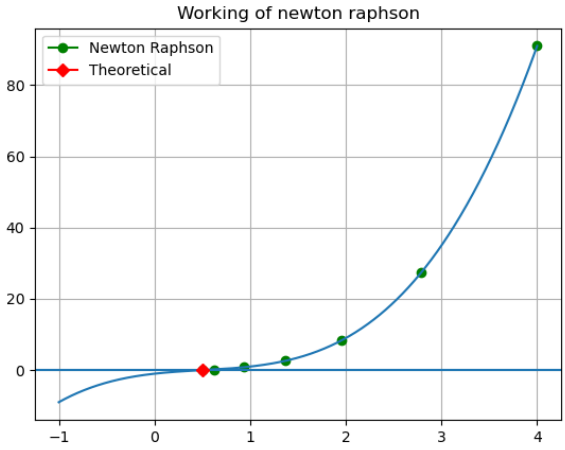


Fig. 3.2.2.1. The depiction of newton raphson method guessing the root

3.2.3 If the matrix $\begin{pmatrix} x & -3 \\ 2 & x-5 \end{pmatrix}$ is singular, sum of the values of x is (AG 2025)

a) 5 b) 6 c) 0 d) -1

Solution: Through row reduction,

$$\begin{pmatrix} x & -3 \\ 2 & x-5 \end{pmatrix} \xleftrightarrow{R_2 \leftrightarrow R_1} \begin{pmatrix} 2 & x-5 \\ x & -3 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow 2R_2 - xR_1} \begin{pmatrix} 2 & x-5 \\ 0 & -6-x^2+5x \end{pmatrix}$$

For the matrix to be singular, from Definition 1.1.23,

$$x^2 - 5x + 6 = 0 \tag{3.2.3.1}$$

Hence, the sum of roots is 5. The following code verifies the result

codes/echelon.py

3.2.4 $\frac{s}{s^2+a^2}$ is the Laplace transform of: (AG 2025)

a) $\cos(at)$
b) $\sin(at)$
c) $\sinh(at)$
d) $\cosh(at)$

Solution: From (3.1.5),

$$\cos(at) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2+a^2} \tag{3.2.4.1}$$

3.2.5 The median of the following set of numbers: 154, 130, 144, 137, 156, 146, 138, 149, 160, 138 is: (AG 2025)

(A) 145.0
(B) 145.2
(C) 151.0
(D) 146.0

Solution: To find the median, we need to arrange the data in ascending or descending order. After arranging them in ascending order, the data is as follows

$$130, 137, 138, 138, 144, 146, 149, 154, 156, 160 \tag{3.2.5.1}$$

The number of elements in the data set are 10. Since it is an even number, the median (M) is defined as the average of the middle 2 terms (in this case, average of 5th and 6th terms). Therefore, the median(M) is $\frac{144+146}{2} = 145.0$. The following codes verify the result:

codes/Sorter.py
codes/Quicksort.c

3.2.6 The series
$\sum_{n=0}^{r} q^n = 1 + q + q^2 + \ldots\ldots\ldots$ has the sum: (AG 2025)

a) $\frac{1}{1-q}$-$\frac{q^{r+1}}{1-q}$
b) $\frac{1}{q-1}$+$\frac{q^{r+1}}{q-1}$
c) $\frac{1}{1-q}$+$\frac{q^{r+1}}{1-q}$
d) $\frac{1}{q-1}$

Solution: The given sequence is a geometric progression with initial term as 1 and common ratio as q and the number of terms are $r+1$.(The series converges only if $q < 1$). Applying formula (3.1.10.1), we get :

$$1\,\frac{q^{r+1}-1}{q-1} \tag{3.2.6.1}$$

$$\frac{1-q^{r+1}}{1-q} \tag{3.2.6.2}$$

$$\frac{1}{1-q} - \frac{q^{r+1}}{1-q} \tag{3.2.6.3}$$

Therefore, the answer is option A. The following code verifies the result:

codes/gp.py

3.2.7 The value of $\lim_{x\to 0} \frac{x\cos(x)-\sin(x)}{x^2\sin(x)}$ (AG 2025)

a) $\frac{-1}{3}$
b) $\frac{1}{3}$

c) 3
d) -3

Solution: We use the idea of taylor series expansion (converges if $x < 1$) to solve the problem.

$$\frac{x\left(\cos\left(x\right)\right) - \sin\left(x\right)}{x^2 \sin\left(x\right)} \tag{3.2.7.1}$$

$$\frac{x(1 - \frac{x^2}{2!}.....) - (x - \frac{x^3}{3!}.....)}{x^2(x - \frac{x^3}{3!}.....)} \qquad \text{(Refer (3.1.11))}$$

When x approaches zero, we can ignore the higher power terms.

$$\frac{(x - \frac{x^3}{2}) - (x - \frac{x^3}{6})}{x^3(1 - \frac{x^2}{6})} \tag{3.2.7.2}$$

$$\frac{\frac{-1}{3}}{1 - \frac{x^2}{6}} \tag{3.2.7.3}$$

Substituting x as zero gives us the limit as $\frac{-1}{3}$.
The following code simulates the result:

codes/Plottinglimit.py

The output graph is shown below:

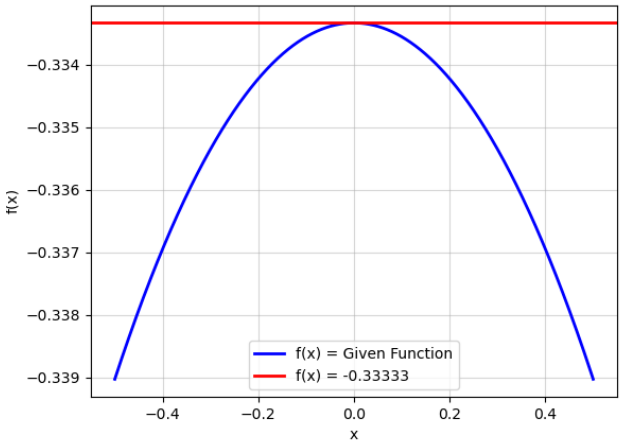


Fig. 3.2.7.1. The plot depicting the approaching of the limit to $\frac{-1}{3}$

3.2.8 If $u = x^4 + y^4 + 3x^2y^2$, the value of $\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)^{-1}$ (AG 2025)

a) $4u$
b) 0
c) $\frac{1}{4u}$
d) $2u$

Solution: We use Euler's theorem for homogeneous functions.(Refer to (3.1.12.1) for the theorem)

$$u = g(x, y) = x^4 + y^4 + 3x^2y^2 \tag{3.2.8.1}$$

$$g(\alpha x, \alpha y) = \alpha^4\left(x^4 + y^4 + 3x^2y^2\right) \tag{3.2.8.2}$$

Therefore, u is a homogeneous function of degree 4. Since we know that the given equation is homogeneous and of degree 4, we can apply this theorem.

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 4u \qquad \text{(Refer equation (3.1.12.1))}$$

Since they asked for its reciprocal, the answer is $\frac{1}{4u}$

3.2.9 The value of y at $x = 0.3$ and $y(x = 0) = 0$ for the differential equation $\frac{dy}{dx} = 3y + 2e^x$ is _____. (Rounded off to 3 decimal places) (AG 2025)

Solution:

a) Let us define a function G(s) such that:

$$y \overset{\mathcal{L}}{\longleftrightarrow} G(s) \tag{3.2.9.1}$$

$$y' \overset{\mathcal{L}}{\longleftrightarrow} sG(s) \qquad \text{(Refer (3.1.13.1))}$$

Applying the laplace transform:

$$sG(s) = 3G(s) + 2\mathcal{L}\{e^x\} \qquad \text{(From (3.1.14))}$$

Now, we evaluate G(s)

$$G(s)(s-3) = \frac{2}{s-1} \tag{3.2.9.2}$$

$$G(s) = \frac{2}{(s-1)(s-3)} \tag{3.2.9.3}$$

Using partial fractions:

$$G(s) = \left(\frac{1}{s-3} - \frac{1}{s-1}\right) \tag{3.2.9.4}$$

Finding y using inverse laplace transform:

$$y \overset{\mathcal{L}}{\longleftrightarrow} G(s) \tag{3.2.9.5}$$

From (3.1.15) and (3.1.16.1),we can write:

$$y = e^{3x} - e^x \tag{3.2.9.6}$$

finding y based on initial conditions:

$$x = 0.3 \quad \text{(given)}$$
$$y = e^{3\times 0.3} - e^{0.3} \quad (3.2.9.7)$$
$$y = 1.110 \quad (3.2.9.8)$$

b) Using forward euler method (see (3.1.17))to find the graph.The difference function of forward euler becomes:

$$y_{n+1} = y_n + h(3y_n + 2e^x) \quad (3.2.9.9)$$

Using it, the following graph is obtained:

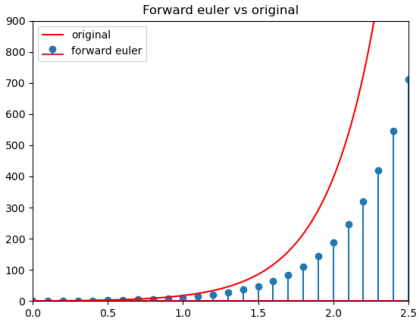


Fig. 3.2.9.1. The plot depicting the divergence of forward euler from original at large values

The following code simulates the result:

codes/diffeq.py